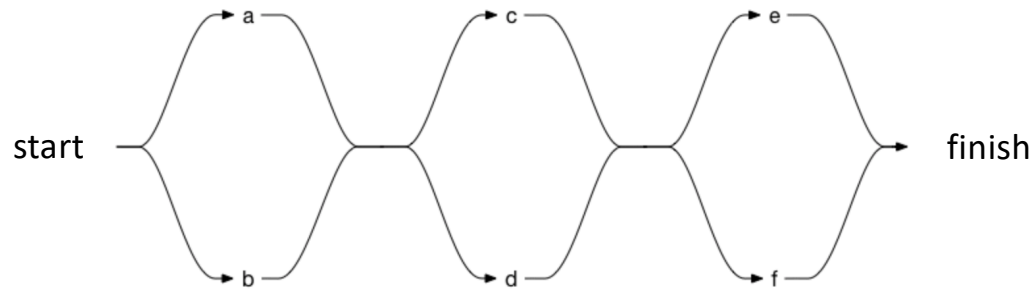


b. The Binomial Theorem

Introduction to binomial theorem

- Suppose we want to evaluate a "product of sums" expression like:
 - $(a + b) \cdot (c + d) \cdot (e + f)$
- We can envision this expression graphically like:



- And evaluating this product corresponds to summing the values of all paths through the circuit from start to finish
- For example: could have first gone through b then d then e
- So $b \cdot d \cdot e$ is a term in the expression, along with all other such paths

Introduction to binomial theorem 2

- Of course we can think of a similar diagram to help us compute an expression like $(x + y)^4$.
 - We would have four consecutive “choices” of x or y
- How many (non-distinct) terms in the final sum?
 - There's $2^4 = 16$ terms in the final sum, corresponding to the total number of paths
- Clearly each path does not correspond to a unique term – and the degree of each term is 4.
- So how many times, for instance, do we get the term x^3y ?
 - This is equivalent to either $C(4,3) = 4$ -- if we count the number of ways to get three factors of x
 - Or we count this by $C(4,1) = 4$ – if we count the number of ways to get a single factor of y
- Notice that $C(n, k) = C(n, n - k)$
 - Prove this by the formula definition of combinations

The Binomial theorem

- More generally we note in the product $(x + y)^k$, we have each term $x^l y^{k-l}$ (or “monomial”) appearing $C(k, l) = C(k, k - l)$ times.
- Thm. For all natural numbers $n > 0$:
 - $(x + y)^n = \sum_{i=0}^n \binom{n}{i} x^{n-i} y^i$
 - Where the notation $\binom{n}{i} = C(n, i)$
 - We’ve already sketched a proof... how would we prove this formally?
- Ex: What is the coefficient of $x^{11} y^5$ in the expansion of $(x + y)^{16}$?
 - Answer: $\binom{16}{11}$
- Ex. What is the coefficient of $x^7 y^4$ in $(x + 2y)^{11}$?
 - Answer: Note without the “2” we would have had $\binom{11}{7}$, but every time we see a y factor we get an extra factor of 2. So answer is $2^4 \binom{11}{7}$

Binomial identities

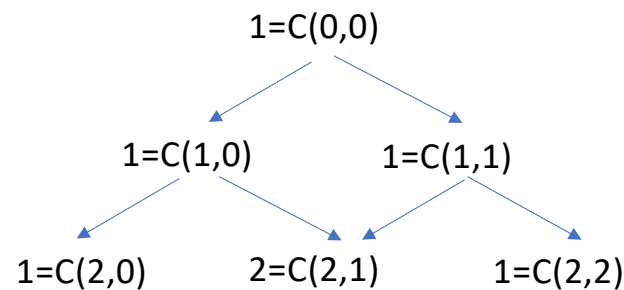
- Recall (binomial theorem): Thm. For all natural numbers $n > 0$:
 - $(x + y)^n = \sum_{i=0}^n \binom{n}{i} x^{n-i} y^i$
- Thm. For all natural numbers $n > 0$, $\sum_{i=0}^n \binom{n}{i} = ?$
- Pf. LHS is exactly the expansion of $2^n = (1 + 1)^n$
- Thm. For all natural numbers $n > 0$, $\sum_{i=0}^n (-1)^i \binom{n}{i} = ?$
- Pf. Consider $0 = (1 - 1)^n$. The binomial theorem says this is exactly equivalent to LHS.

Pascal's identity

- Thm. For all natural numbers $n \geq k > 0$, $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$
- Pf (*sketch*). The LHS is considering an $n+1$ element set S and taking the number of k -element subsets. We could equivalently count this by considering:
 - A fixed element $s \in S$ and a k -element subset of S called K
 - Now there are two disjoint possibilities:
 - Either $s \in K$
 - So we know that $k - 1$ elements of K come from set $S - \{s\}$
 - Notice $S - \{s\}$ is a set of size n , and we're interested in a subset of size $k - 1$
 - So $\binom{n}{k-1}$ ways to choose this
 - Or $s \notin K$
 - In which case all k elements of K come from $S - \{s\}$
 - As before $S - \{s\}$ is a set of size n , but now we're interested in a subset of size k
 - So $\binom{n}{k}$ ways to choose

Pascal's triangle

- Recall the identity: $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$
- We can visualize the binomial coefficients in the following geometric way:
 - A “triangle” in which the k-th row (starts with 0) has entries $C(k,0) \dots C(k,k)$
 - Notice, by Pascal's identity, we have the following relations (pair of arrows indicate sum)



- Where, for example, in third row (i.e., $n+1=2$, $k=1$) we have $C(2,1)=C(1,0)+C(1,1)$
- And so on...

Generalization: Vandermonde's Identity

- Thm. For all $m, n \geq r > 0$, $\binom{m+n}{r} = \sum_{k=0}^r \binom{m}{r-k} \binom{n}{k}$
- Pf. Similar idea as before...
 - Let A, B be disjoint sets of size m, n .
 - LHS counts the number of subsets of size r of $A \cup B$
 - But there's another way to count this...
 - Can also count this by counting for distinct $0 \leq k \leq r$, the number of ways to select a subset of size $r - k$ from A and a subset of size k from B
- Cor. $\binom{2n}{n} = \sum_{k=0}^n \binom{n}{k} \binom{n}{k}$
- Pf. In above equality, consider two equal sized sets, so let $m = r = n$ and appeal to symmetry (recall: $\binom{n}{k} = \binom{n}{n-k}$)

Final binomial identity

- Thm. Let $n \geq r > 0$. Then $\binom{n+1}{r+1} = \sum_{j=r}^n \binom{j}{r}$
- Pf. LHS is way to pick $r+1$ elements from set S , of size $n+1$.
 - There's another way to do it!
 - To simplify, let's say the elements of S are $\{0,1,2,\dots,n\}$
 - What is the *minimal* greatest element a subset of S of size $r+1$ could have?
 - It's r
 - How many subsets of S of size $r+1$ have greatest element r ?
 - It's $C(r,r)=1$
 - In general, the number of subsets of S that have the largest element $j \geq r$ is $C(j,r)$ -- why?
 - We're effectively fixing a single element, j , of the subset (the largest element)
 - Since the subset is of size $r+1$, we have r left to choose from $0,\dots,j-1$